

Design of an Autonomous Drone Inspection System for Transmission Lines

Zhen Chen ^{1,*}Jun Li ¹¹ Liaoning Power Transmission and Transformation Engineering Co., Ltd.¹ Liaoning Power Transmission and Transformation Engineering Co., Ltd.

Shenyang 110021, Liaoning, China

Shenyang 110021, Liaoning, China

*Corresponding author's e-mail: gjdwsbd@sina.com

Ming Dong ¹Longping He ¹¹ Liaoning Power Transmission and Transformation Engineering Co., Ltd.¹ Liaoning Power Transmission and Transformation Engineering Co., Ltd.

Shenyang 110021, Liaoning, China

Shenyang 110021, Liaoning, China

Abstract—To address the numerous challenges encountered in the operation and maintenance of 500kV transmission lines, this paper proposes the development of an autonomous drone inspection system for transmission lines. Adopting a three-tier architecture comprising terminal execution, edge collaboration, and cloud management, the system achieves autonomous transmission line inspection through integrated smart poles. It incorporates four functional modules: an adjustable fixed drone nest, photovoltaic storage and charging, image and data transmission, and anti-theft security. System stability is ensured through nest elevation mechanisms, adaptive tower design, photovoltaic energy storage power supply, dual-link transmission, and anti-theft security measures. A complementary remote management platform enables task scheduling, operational monitoring, and AI analysis. During the pilot deployment on Shenyang's 500kV Puxi Lines 1 and 2, two drone nest systems covered 23 pylons (8.3 kilometres), expanding the operational radius by over 50%. Fault response times were significantly reduced, maintaining stable power supply throughout ten consecutive days of adverse weather conditions. This provides a reliable solution for high-voltage line maintenance.

Keywords: Drone inspection; Smart Pole; Machine Nest; Photovoltaic storage system

I. INTRODUCTION

With increasing demands for power supply across all sectors, ensuring the reliability of electricity provision has become a crucial foundation for the stable operation of the national economy. As the backbone channels for power transmission, high-voltage transmission lines necessitate regular inspection and maintenance ^[1]. According to operational maintenance data from State Grid Liaoning Electric Power Co., Ltd., the current maintenance of 500kV transmission lines faces challenges including insufficient deployment of proprietary unmanned aerial vehicles (UAVs) and a limited pool of certified UAV operators. Furthermore, manual inspections in complex terrain such as mountainous and forested areas present drawbacks including heightened safety risks, numerous blind spots, and slow response times. Manual inspections in complex terrain such as mountainous areas also present drawbacks, including

heightened safety risks and slow response times^[2]. Although UAV inspection technology has been applied in power grid operations, existing UAV airfields predominantly rely on fixed deployment points such as substations and supply stations. Simultaneously, field deployment faces challenges including lack of mains power supply, susceptibility to signal transmission blockage and interference, and inadequate equipment security. Therefore, this paper develops an autonomous drone inspection system capable of flexible deployment along transmission lines. The system features an elevatable nest design^[3], autonomous photovoltaic energy storage, signal amplification, and tower platform adaptability. By overcoming the constraints of fixed locations, the nest deploys along transmission lines, maximising the operational radius per station and significantly expanding inspection coverage. With self-sufficient energy supply and stable signal transmission capabilities, it effectively addresses current pain points in transmission line inspection ^[4].

II. OVERALL DESIGN OF AN AUTONOMOUS UAV INSPECTION SYSTEM FOR TRANSMISSION LINES

A. System Architecture

The system architecture comprises three layers: the terminal execution layer, the edge coordination layer, and the cloud management layer. Each layer achieves bidirectional interaction through data transmission chains and control command chains. The terminal execution layer centres on autonomous smart poles for power line inspection as core field equipment, integrating four functional modules: adjustable fixed drone nests, photovoltaic energy storage charging, image data transmission, and anti-theft security. This enables flexible deployment along transmission tower routes. The edge coordination layer is deployed at regional operations centres. Leveraging localised data processing and task scheduling technologies, it primarily addresses challenges of long-distance data transmission latency and the coordinated dispatch of multiple drone nests. The cloud management layer establishes a dedicated intelligent drone inspection cloud platform, providing functions such as task management, data storage, and AI analysis, whilst supporting **integration with the State Grid's unified management platform.**

State Grid Corporation of China Science and Technology Project Funding (Contract No.: 2025ZX-15)

B. Core Objective

The core design objectives of this autonomous UAV inspection system for transmission lines focus on multi-dimensional technical performance: - Deployment and elevation: The nest must ascend to a minimum height of 8 metres, maintain an operational clearance of no less than 5 metres, and descend to 1.2 – 3 metres during maintenance with smooth, vibration-free elevation. Deployment along transmission towers must cover 10 – 13 structures. Regarding energy supply, the photovoltaic storage system must sustain continuous operation for 10 consecutive overcast or rainy days. Under optimal sunlight conditions, it shall support an average of 6 or more daily flight operations, and 3 or more operations during low-light conditions. Data transmission must maintain a packet loss rate not exceeding 1% for image and data transmission, support real-time transmission of high-definition 1080P imagery, and provide effective coverage with a radius of no less than 10km. It must be adaptable to corresponding operating temperatures and relative humidity, with a protection rating no lower than IP65. In terms of automation, the time from task assignment to drone take-off shall be under 3 minutes, the time from task completion to charging and subsequent standby mode shall be under 2 minutes, and it shall operate continuously without failure for over 180 days.

III. TECHNICAL DESIGN OF CORE SYTEM MODULES

A. Elevating fixed machine nests and tower deployment technology

The lifting mechanism for the drone nest (as illustrated in Figure 1) employs an electric servo drive and gear-meshing transmission solution. Its lifting drive system incorporates a 1000W servo motor, paired with a precision gear-meshing transmission mechanism featuring a 1:20 gear ratio, ensuring smooth and vibration-free operation during elevation. Additionally, the structure incorporates comprehensive safety locking and protective mechanisms. Mechanical limit pins are installed at critical positions along the lifting rod (1.2m, 3m, 5m, 8m), complemented by electromagnetic locking devices to prevent accidental falls during operation. An infrared beam sensor is integrated within the lifting mechanism; upon detecting an obstacle, it automatically halts and reverses direction to avert damage to equipment or personnel^[5].



Figure 1. Airport Elevator Mechanism

The tower platform employs a clamp-type connection with a high-strength, lightweight structure. The main frame is constructed from Q420 high-strength hot-dip galvanised steel, coated with heavy-duty anti-corrosion paint exceeding 150 μ m thickness to meet long-term outdoor corrosion resistance requirements. Platform dimensions are 1200mm $\times$ 1200mm $\times$

800mm, with a weight under 35kg (excluding UAV and power supply), compatible with installation spaces on the first or second platform of 500kV transmission towers. The drone platform is mounted at the cross-braced section of the tower legs, secured to the main tower structure (angle steel) via two sets of M24 high-strength clamps measuring 100mm in width and 12mm in thickness. This eliminates the need for additional drilling, thereby preventing structural weakening of the tower (The installation diagram is shown in Figure 2). Verified through finite element analysis using ANSYS software (as illustrated in Figure 3), the platform can withstand combined loads comprising the drone housing (37kg), photovoltaic storage power supply (500kg), maintenance personnel (160kg), and wind speeds up to Beaufort scale 8 (wind pressure 0.5kN/m²). The maximum stress ratio remains below 0.7, satisfying the structural safety margin requirements for transmission towers.



Figure 2. Schematic Diagram of Drone Nest Installation

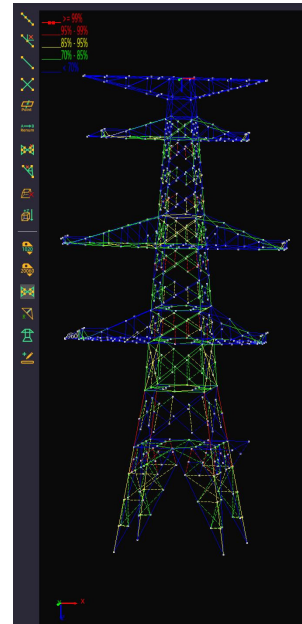


Figure 3. Stress Ratio Contour Map Of The Tower

B. Photovoltaic storage and charging system design

For scenarios without mains electricity in the field, the photovoltaic-storage system employs a photovoltaic, lithium-ion battery, and intelligent control architecture^[6]. This specifically comprises four 18V/180W series-connected monocrystalline silicon photovoltaic panels, a 12.8V/250Ah lithium iron phosphate battery pack, and a 1500W MPPT inverter-charger unit (featuring 4G remote monitoring, enabling real-time energy data upload and supporting remote operation

and maintenance). Additionally, the system employs deep sleep and dynamic scheduling energy management strategies. During idle periods, non-essential equipment power is disconnected, retaining only 4G heartbeat and temperature sensors. Power is restored within three minutes of receiving remote tasks for rapid response, whilst operational frequency dynamically adjusts based on light intensity and task requirements.

C. Design of image and data transmission systems

To address signal obstruction and electromagnetic interference in mountainous and densely populated areas, the system employs high-gain antennas and a dual-link transmission solution. Dual-band high-gain antennas operating at 2400-2500MHz and 5150-5850MHz are installed atop autonomous transmission line inspection smart poles, elevated to a height of 5 metres to minimise ground obstruction. This configuration achieves an effective coverage radius exceeding 10 kilometres. Simultaneously supporting parallel operation of 4G/5G public networks and a digital image transmission module with a control range of no less than 15km. The public network link ensures remote control and data backhaul, while the digital image transmission module serves as a backup to prevent mission failure due to single-link disruption. To guarantee transmission stability, the protocol employs a UDP and RTCP combination: UDP ensures rapid delivery of 1080P HD imagery, while RTCP monitors transmission quality in real-time, automatically triggering link switching when packet loss exceeds 1%. The digital image transmission module features a metal-shielded housing to minimise electromagnetic interference.

To validate the communication stability of the dual-link transmission system across varying distances and environments, multi-scenario testing was conducted (with an 80% tree canopy cover in densely wooded areas and a signal interference zone selected near a 220kV substation). Specific data is presented in the table below.

TABLE I. TRANSMISSION PERFORMANCE COMPARISON

Testing Scenario	Testing Distance	Transmission Status of Single-link	Transmission Status of Dual-link
Open Plain	5km	Packet loss rate 0.3%	Packet loss rate 0.2%
Open Plain	10km	Packet loss rate 1.8%	Packet loss rate 0.5%
Dense Tree Area	5km	Packet loss rate 2.1%	Packet loss rate 0.4%
Dense Tree Area	10km	Packet loss rate 5.3%	Packet loss rate 1.0%
Signal Interference Area	5km	Packet loss rate 3.2%	Packet loss rate 0.6%
Signal Interference Area	10km	Packet loss rate 7.8%	Packet loss rate 1.2%

D. Design of anti-theft security and environmental adaptation systems

To safeguard on-site equipment, this unit incorporates a 360-degree panoramic 1080P night vision camera (featuring motion

detection, night vision infrared imaging, and real-time image transmission to a remote management platform). The system integrates vibration sensors and door magnetic sensors (triggering immediate audible and visual alarms while alerting maintenance personnel upon detecting abnormal vibrations or unauthorised door openings). Additionally, both the battery compartment and control cabinet feature dedicated anti-theft locks. The housing employs stainless steel exceeding 2 millimetres in thickness, achieving an IP65 protection rating to effectively prevent dust ingress, water ingress, and physical damage. To withstand harsh outdoor conditions, the system incorporates temperature, humidity, and anti-corrosion measures for enhanced adaptability: The internal PT01 temperature sensor (as shown in Figure 4) continuously monitors the battery charging compartment temperature. This sensor utilises a compact, high-temperature-resistant DS18B20 armoured digital sensor. Its installation location must avoid all moving component areas, including the hatch opening trajectory, drone centring path, and the area directly above the charging contacts. Thermal grease is applied to enhance temperature response speed and accuracy. All metal components undergo hot-dip galvanisation to withstand prolonged outdoor corrosion. Silicone rubber seals secure the compartment doors. Spring washers and lock nuts jointly prevent loosening at tower platform connection points.



Figure 4. Engine Bay Temperature Sensor

E. Design of integrated smart poles for autonomous transmission line inspection

As the core carrier deployed along transmission towers, the smart pole integrates equipment housing support, antenna mounting, and photovoltaic storage fixation capabilities, resolving limitations of traditional deployments. Its pole body adopts a segmented structure (base section 400×400mm, middle section 200×200mm, top section 100×100mm) with a total height of 8m. Flange connections and a weight under 150kg facilitate transportation and installation. Additionally, the pole incorporates pre-installed interfaces for equipment pods at mid-section, accommodates high-gain antennas at the top, and secures photovoltaic storage battery boxes and control cabinets at the base. This multi-functional design reduces additional loads on transmission towers. The pole also interfaces with the equipment pod lifting system: during operations, the pole maintains a fixed height, while during maintenance, the pods descend with the pole to 1.2 metres, streamlining operational and maintenance procedures.

IV. REMOTE CONTROL PLATFORM DESIGN

This remote control platform employs a three-tier architecture comprising cloud, edge and terminal components,

enabling centralised management and intelligent dispatch of multiple drone nests and unmanned aerial vehicles. The system incorporates a dual-link redundancy mechanism within its communication transmission design, effectively addressing scenarios such as densely forested areas and regions with weak signals, thereby providing robust assurance for inspection data transmission and command interaction. The task management module utilises drones to conduct LiDAR scanning of ultra-high voltage transmission lines spanning 30 kilometres^[7]. By collecting laser data to construct three-dimensional models, it establishes a database of tower coordinates and transmission line models (delivered data includes voltage level, line name, tower latitude/longitude, point cloud density ≥ 100 points/square metre, absolute accuracy error ≤ 20 centimetres, coverage extending 10 metres on either side of the line conductor, with no omission of key points). This enables transmission of line data (as illustrated in Figure 5). The system automatically generates inspection flight paths on GIS maps (with at least three photographic points designated per tower to ensure comprehensive coverage of critical components). It supports both scheduled and emergency tasks, dynamically allocating missions based on drone nest energy reserves and drone battery status to achieve leapfrog inspections. The real-time monitoring module tracks task progress in real time, displaying the current status of the drone nest, drones, and photovoltaic energy storage systems (with abnormal data highlighted in red alerts). It also supports multi-channel first-person view live streaming from drones, allowing switching between visible light and infrared thermal imaging modes. The data analysis module integrates an artificial intelligence defect recognition algorithm based on the ResNet50 network. It automatically analyses inspection images to identify faults and generate reports, while consolidating relevant statistical data into operational maintenance reports to provide decision support for line maintenance^[8].

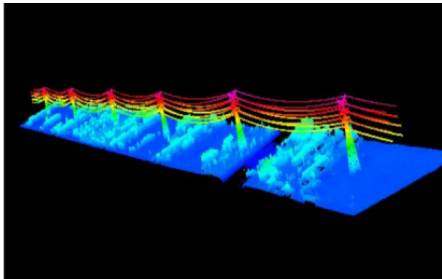


Figure 5. Three-dimensional Point Cloud Imaging Diagram

V. SYSTEM PILOT APPLICATION AND EFFECTIVENESS VERIFICATION

The Shenyang 500kV Puxi No.1 and No.2 transmission lines were selected as the pilot project. These lines traverse hilly terrain with complex environmental conditions, including tree obstructions and RC signal interference. The deployment environment is illustrated in Figure 6. Two adjustable fixed nests were deployed (installed on towers #080 and #093 respectively), covering 23 towers from #077 to #099. The total line length is 8.3km, as illustrated in Figure 7.



Figure 6. Surrounding Environment



Figure 7. Airport Coverage Map

The antenna pod is secured to the tower via clamps without compromising the tower structure. A single pod can cover the #077-#086 (10 towers, 4.6km) and #087-#099 (13 towers, 3.7km) sections. Its operational radius exceeds substation deployments by over 50%, whilst high-gain antennas and dual-link transmission effectively mitigate signal interference. As illustrated in Figures 8-9: - Tower #077 (1.9km from #080) exhibits normal 4G video transmission signals, with enhanced RC signal interference detected during testing; - Tower #086 (2.7km from #080) maintains normal 4G video transmission signals with a 0.8% packet loss rate, meeting high-definition image transmission requirements. As illustrated in Figures 10-11: - At test height 109 metres (2.1km from Tower #087), RC signal exhibits slight interference while 4G video transmission remains unaffected by interference during flight operations. - At test height 183 metres (1.6km from Tower #087), RC signal shows slight interference but 4G video transmission remains interference-free.

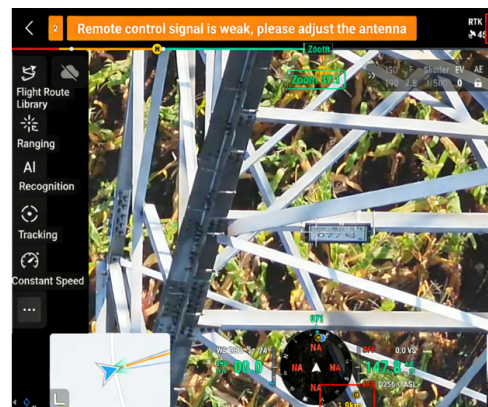


Figure 8. #077 Signal Test Chart

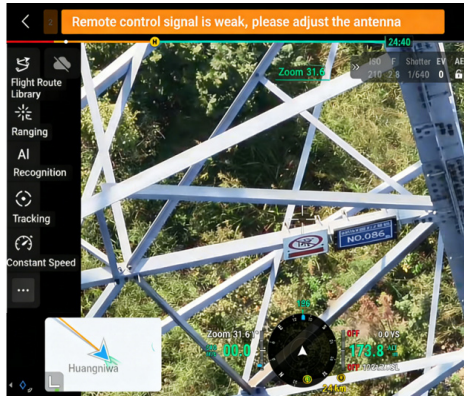


Figure 9. #086 Signal Test Chart

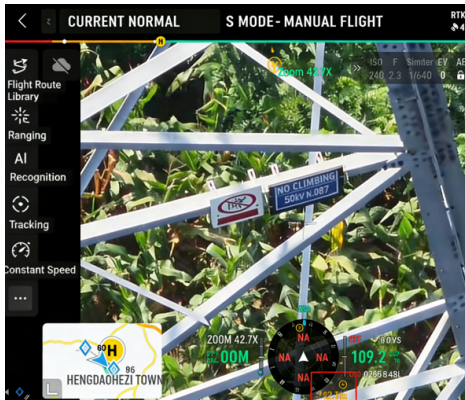


Figure 10. #087 Signal Test Chart

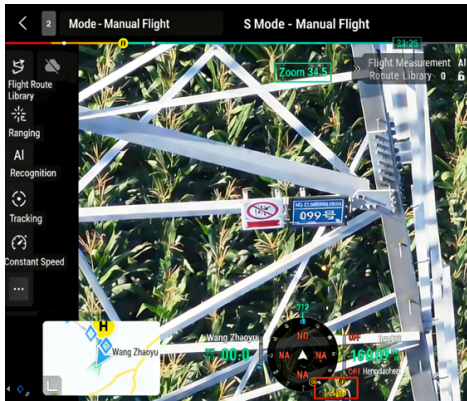


Figure 11. #099 Signal Test Chart

To systematically validate the long-term operational reliability and performance stability of the autonomous drone inspection system for transmission lines under extreme weather conditions, a 30-day specialised monitoring experiment was conducted on the pilot 500kV Puxi 1 and 2 transmission lines in Shenyang (covering the #077-#099 tower section). The dual-link transmission system (4G/5G public network + digital video transmission) dynamically adjusted transmission parameters via the RTCP real-time transmission quality monitoring protocol when signal attenuation occurred due to heavy rainfall. When the 4G public network packet loss rate exceeded 1%, it automatically switched to the digital video transmission module

(operating at a distance of 15km). During monitoring, the average data transmission packet loss rate was 0.62%, with a peak of 0.8% occurring during heavy rainfall on Day 5. No image stuttering or data loss was observed. The AI defect recognition algorithm achieved 92.3% accuracy in identifying faults such as insulator damage and conductor water coverage. This represents a deviation of only 0.7% from the 93% accuracy recorded under clear weather conditions, ensuring robust data validity.

In terms of energy and operations, the photovoltaic storage system generates an average of 4.2 kWh daily during summer, supporting six inspection flights. Daily power consumption averages 1.47 kWh. During winter, with daily generation adjusted to 2.5 kWh, operations were scaled to one flight every two days. No power interruptions occurred over ten consecutive overcast days. Upon dispatching fault inspection tasks, the drone nest completed wake-up and launch within 2 minutes 30 seconds, reducing response time by 97% compared to manual drone deployment (average 2 hours). Regarding safety and operations, no incidents of equipment theft or vandalism occurred during the pilot phase. Anti-theft cameras successfully identified two instances of unauthorised personnel approaching and triggered alerts, prompting timely intervention by maintenance personnel. Additionally, the drone nest can be lowered to 1.2 metres during maintenance, reducing the maintenance time per unit from the traditional four hours required for high-altitude operations to just one hour, thereby lowering operational costs by 60%.

VI. CONCLUSION

The autonomous drone inspection system for transmission lines designed in this study addresses numerous limitations of conventional inspection methods. Supported by a three-tier architecture, the terminal execution layer enables flexible deployment along pylons and multifunctional integration; the edge coordination layer optimises regional task scheduling and data pre-processing; while the cloud management layer achieves comprehensive intelligent oversight. These tiers interact efficiently through data transmission and control command chains. Pilot application validation demonstrates the system's remarkable efficacy in deployment coverage, energy self-sufficiency, response velocity, and secure operations. It substantially enhances both the inspection efficiency and coverage of 500kV transmission lines while significantly reducing reliance on manual labour and operational costs. Furthermore, it shortens fault response times, providing robust support for the power industry's transition from labour-intensive to technology-intensive inspection models. The system thus possesses extensive value for widespread implementation.

REFERENCES

- [1] Shenyang. Application Strategies for Drones in Power Equipment Fault Detection [J]. *Lighting and Illumination*, 2025, 49(05): 142-144.
- [2] Nie, Houyong. Research and Design of a High-Voltage Transmission Line Inspection System Based on Unmanned Aerial Vehicles [Doctoral Dissertation]. Hefei University of Technology, 2024. DOI: 10.27101/d.cnki.ghfgu.2024.000881.
- [3] Qiu Linfeng, Huang Zijun, Pang Meijun. Research and Design of an AI-Based Intelligent Inspection, Monitoring and Early Warning Platform for Broadcasting and Television Networks Utilising Drone Nests [J].

- Broadcasting and Television Network, 2025, 32(02): 35-38. DOI: 10.16045/j.cnki.catvtec.2025.02.010.
- [4] Hu Hongwei, Li Ming, Cheng Junxiang, et al. Research on an All-Weather Autonomous Inspection Drone Nest System for Transmission Lines [J]. Northeast Electric Power Technology, 2025, 46(09): 27-36.
 - [5] Maijun Jia, Cui Wei, Zeng Yihui. Nest Configuration Deployment Method for Transmission Line Inspection Drones [J]. Guangdong Electric Power, 2023, 36(02): 102-108.
 - [6] Ma Zheng, Lan Lu. Technical Research on Wireless Charging Systems for Unmanned Aerial Vehicles [J]. Shanxi Electric Power, 2025, (04): 24-27.
 - [7] Wang Feiran, Guo Xinyang, Han Geng, et al. Digital Analysis of Clearance Safety for Overhead Transmission Lines Driven by UAV Point Clouds [J]. China New Technology and New Products, 2025, (18): 33-37. DOI: 10.13612/j.cnki.cntp.2025.18.011.
 - [8] Gong Jiahua. Research on UAV Inspection Technology for 500kV Smart Substations [J]. Electrical Technology and Economics, 2025, (10): 67-69.